BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis (MAT221)

Lebesgue's Criterion for Integrability

Measure Theory, Lebesgue's Theorem

September 06, 2026

CONDUCTED BY

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

partho.dhor@bracu.ac.bd | parthosutradhor@gmail.com

For updates subscribe on @ParthoSutraDhor

Sets of Measure Zero

Theorem

A set $A \subseteq \mathbb{R}$ is said to be of measure zero if for every $\epsilon > 0$, there exists a countable collection of open intervals $\{U_n\}_{n=1}^{\infty}$ such that

$$A \subseteq \bigcup_{n=1}^{\infty} U_n \text{ and } \sum_{n=1}^{\infty} \mu(U_n) < \epsilon,$$

where $\mu(U_n)$ denotes the measure of the interval U_n.

Countable Sets are of Measure Zero

Every countable set $A \subseteq \mathbb{R}$ is of measure zero.

Continuity and Oscillation

Oscillation of a Function

Let $f : [a, b] \to \mathbb{R}$ be a bounded function. The oscillation of f at a point $x_0 \in [a, b]$ is defined as

$$\omega_f(x_0) = \inf_{\delta>0} \left(\sup_{|x-x_0|<\delta} f(x) - \inf_{|x-x_0|<\delta} f(x) \right).$$

Theorem

Let $f : [a, b] \to \mathbb{R}$ be a bounded function. Then f is continuous at a point $x_0 \in [a, b]$ if and only if the oscillation of f at x_0 is zero, i.e., $\omega_f(x_0) = 0$, where

$$\omega_f(x_0) = \inf_{\delta>0} \left(\sup_{|x-x_0|<\delta} f(x) - \inf_{|x-x_0|<\delta} f(x) \right)$$

Lebesgue's Theorem

Theorem

A bounded function $f : [a, b] \to \mathbb{R}$ is Riemann integrable if and only if the set of points where f is discontinuous has measure zero.

Thank You!

We'd love your questions and feedback.

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

partho.dhor@bracu.ac.bd | parthosutradhor@gmail.com

@ParthoSutraDhor

(Lectures, walkthroughs, and course updates)

Scan for the channel

References

[1] Stephen Abbott, *Understanding Analysis*, 2nd Edition, Springer, 2015.

[2] Terence Tao, *Analysis I*, 3rd Edition, Texts and Readings in Mathematics, Hindustan Book Agency, 2016.